**Semester VII** **B.Tech CSE (AI)**

## MBA 204: RESEARCH METHODOLOGY

**2L+0T+0P+2C** **MM:100**

**Unit 1: Meaning And Importance Of Research**
Review of Literature, Objectives of the research, Types of Research- Exploratory Research – Descriptive Research – Casual Research - Research Approaches- Research Process ––Defining Research Problem- Selection and necessity of defining the problem.

**Unit 2: Research Design**
meaning, need and features of good research design- Important concepts related to research design. Experimental research designs: Before and After without control design, After only with control design, Before and after with control design, Completely randomized design (C.R Design).
**Sampling and Sampling Design** – Sampling Methods – Simple Random Sampling – Stratified Sampling – Systematic Sampling – Cluster Sampling – Multistage Sampling, Non–Probability Sampling – Convenience Sampling – Judgment Sampling – Quota Sampling- Snowball sampling.

**Unit 3**: **Data Collection**
Primary and Secondary Data – Designing of Questionnaire –**Measurement and Scaling** – Nominal Scale – Ordinal Scale – Interval Scale – Ratio Scale –Guttman Scale – Likert Scale – Schematic Differential Scale. Descriptive statistics- Measures of central tendency- Dispersion- Skewness -Correlation and Regression Analysis.

**Unit 4**: **Editing**
Coding – Classification of Data – Tables and Graphic Presentation –Basics of inferential statistics- Types of Errors- **Hypothesis testing** -Parametric test - T-test, Z test, Chi Square test- ANOVA Test. Introduction of SPSS.

**Unit 5** : **Non Parametric Tests**
Kolmogorov – Smirnov Test – Runs Test for Randomness. Sign Test – Median Test –Factor Analysis.
**Preparation and Presentation of Research Report**- Types of reports- Layout of Research Report- Bibliography-References writing- Precautions for writing Research Report.

**Text/ Reference Books:**

1. Mark Saunders, Philip Lewis, Adrian Thornbill, Research Methods for Business Students, Pearson,ND
2. Churchill, Iacobucci & Israel, Marketing Research: A South Asian Perspective, Cengage, New Delhi
3. C.R. Kothari, Research Methodology, New Age International.
4. Carver & Nash, Data Analysis with SPSS, Cengage, New Delhi
5. Alan Bryman & Emma Bell, Business Research Methods, Oxford University Press.
6. Donald R. Cooper & Pamela S. Schindler, Business Research Methods 8th Edition, Tata McGraw Hill.
7. K.V.S. Sarma, Statistics made sample, do it yourself on PC, Prentice Hall.
8. V P Michael, Research Methodology in Management, Himalaya, Mumbai